

**SIMATS ENGINEERING**  
**ASSESSMENT TOOL 1: Short Answer Test**

**COURSE NAME:** Cloud Computing and Big Data Analytics      **COURSE CODE:** CSA1501

---

**COURSE OUTCOME COVERED**

**CO1:** *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)  
Question Count: 5

**Total Marks: 25**

---

**Code of Conduct**      19251171

I H-Harshwardhan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: H-Harshwardhan

**Assessment Tasks**

**Short Answer Test**

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

**Short Answer Test Rubric**

| Criteria                     | Weightage | Level 4 - Exemplary                                           | Level 3 - Proficient                          | Level 2 - Developing                         | Level 1 - Beginning                 |
|------------------------------|-----------|---------------------------------------------------------------|-----------------------------------------------|----------------------------------------------|-------------------------------------|
| Conceptual Accuracy          | 30%       | Accurate definitions and precise explanation of all key terms | Mostly accurate with minor gaps               | Partial understanding with noticeable errors | Incorrect or irrelevant explanation |
| Coverage of Concepts         | 20%       | Covers all expected sub-points completely                     | Covers most expected points                   | Covers only basic points                     | Very limited coverage               |
| Use of Examples              | 15%       | Uses highly relevant cloud examples to strengthen the answer  | Uses relevant examples with minor mismatch    | Uses vague or partially relevant examples    | No useful examples                  |
| Comparison / Differentiation | 20%       | Clearly distinguishes concepts with strong logic              | Reasonable differentiation with minor overlap | Comparison is weak or incomplete             | Fails to differentiate concepts     |
| Clarity of Expression        | 15%       | Clear, organized, and concise answer writing                  | Generally clear with few issues               | Understandable but poorly organized          | Difficult to follow                 |

1) Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

### Definition

Cloud computing is a technology that provides computing resources such as servers, storage, database, networking and software through the internet on a demand basis.

### Characteristics

#### On demand and self services:

User can access computing resources when required without human interaction with service provider.

#### Broad Network access:

Cloud services can be accessed from anywhere using the internet through computers, mobiles and other devices.

#### Resource Pooling:

Cloud provides shared computing and among multiple users using virtualization technology.

#### Resource Elasticity:

Resource can be increased or decreased quickly based on user requirements.

## Difference from Traditional Computing:

Traditional Computing Requires Physical Hardware and the maintenance while Cloud Computing provides flexible resource over the internet.

- 2) Trace the Evolution of Cloud Computing from traditional Evolution to internet Software Evolution and show how these changes Enabled modern Cloud Platforms.

## Evolution of Cloud Computing

### Hardware Evolution

Early system used large mainframe computers. Personal Computers and powerful servers computing capabilities.

virtualization allowed multiple virtual machines on one physical sharing.

### Distributed System:

→ Computing Resources were connected across multiple system technologies like Grid Computing and Cluster Computing improved resource sharing.

### internet Evolution:

→ Growth of high speed internet Enabled remote access to computing resources



\* web Technologies allowed applications to be delivered Online

### Internet software Evolution,

web Applications, APIs, and Service oriented architecture Enabled Online services.

How it enabled modern Cloud Platforms:

- These development created scaled Platforms like Public, Private and hybrid Clouds.
- services such as IaaS, PaaS, and SaaS became possible.

3) Explain the role of server virtualization in Cloud Computing. Differentiate between virtualization as an Enabling technology and cloud Computing as a service model.

Ans: Role of server virtualization

- server virtualization divides one Physical server into multiple virtual machines
- It improves resource virtualization and reduces hardware cost
- It provides Flexibility, Scalability and Efficiency management of cloud resources.

## Difference

| virtualization                                        | cloud computing                                                     |
|-------------------------------------------------------|---------------------------------------------------------------------|
| → It is a technology that creates virtual resources   | → It is a service model that delivers resource through the internet |
| → Uses Hypervisor to create virtual machines          | → Provides services like IaaS, SaaS, PaaS                           |
| → Focus on resource management<br>eg: VMware, Hyper-V | → Gives non-service delivery to user<br>eg: AWS, Microsoft          |

4) Discuss the role of data centers in delivering cloud services

### Role of Data centers:

#### 1. Resource Hosting:

→ Data centers contain servers storage devices and Networking Equipment Required for cloud services.

### Data Storage:

→ The Data Storage and large amount of user data security.

## → Service Delivery

They provide applications, databases and compute resources through the internet

## High availability:

Backup system and redundancy ensure continuous services availability.

## Security management:

Data Centers use fire encryption and monitoring system to protect data.

5) Differentiate IaaS, PaaS, SaaS, XaaS with suitable real world Example for each model.

| model | Description                                                        | Provides responsibility       | Use Responsibility      |
|-------|--------------------------------------------------------------------|-------------------------------|-------------------------|
| IaaS  | Provides virtual Infrastructure like servers, storage and networks | hardware, storage networking  | OS, applications data   |
| PaaS  | Provides a Platform and deploying                                  | Infrastructure, tools.        | Application development |
| SaaS  | Provides ready to use software app over the internet               | Applications, update security | Using Software          |
| XaaS  | Provides any 2+ services through the cloud.                        | Complete service management   | Using required services |